THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A force is a: 1

(a) type of energy only **(b)** kind of speed
(c) state of rest **(d)** push or a pull

2. Pressure is defined as: 1

(a) area per unit force **(b)** mass per unit volume
(c) force multiplied by area **(d)** force acting per unit area

3. According to the particulate model, matter is made up of: 1

(a) tiny particles **(b)** continuous jelly-like material
(c) only air **(d)** only liquids

4. A pure substance that cannot be broken down into simpler substances by chemical methods is a(n): 1

(a) solution **(b)** compound **(c)** element **(d)** mixture

5. A homogeneous mixture of two or more substances is called a: 1

(a) solution **(b)** compound **(c)** element **(d)** suspension

6. Friction is greater between: 1

(a) smooth surfaces **(b)** oiled surfaces
(c) rough surfaces **(d)** polished surfaces

7. The instrument used to measure atmospheric pressure is a: 1

(a) hygrometer **(b)** thermometer **(c)** barometer **(d)** speedometer

8. The change from solid to liquid is called: 1

(a) melting **(b)** boiling **(c)** condensation **(d)** freezing

9. The property of metals by which they can be beaten into thin sheets is: 1

(a) ductility **(b)** sonority **(c)** brittleness **(d)** malleability

10. Which of these solutions is a solid in a solid? 1

(a) soda water **(b)** brass **(c)** air **(d)** vinegar

11. A body is charged by rubbing. The two types of charges are: 1

(a) positive and negative **(b)** heavy and light
(c) hot and cold **(d)** north and south

12. A cyclone has at its centre a region of: 1

(a) very low pressure **(b)** solid ice
(c) very high pressure **(d)** no air

13. Which of the following increases the rate of evaporation? 1

(a) increase in humidity **(b)** absence of wind
(c) increase in temperature **(d)** decrease in surface area

14. A mixture is: 1

(a) a combination of two or more substances not chemically combined
(b) always a liquid
(c) a single compound
(d) a pure substance

15. The mass percentage of a solution is: 1

(a) (mass of solute / mass of solution) x 100
(b) (mass of solution / mass of solute) x 100
(c) (mass of solvent / mass of solute) x 100
(d) mass of solute x mass of solvent

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is a force? State its SI unit. 2

17. Define pressure and write its formula and SI unit. 2

18. State two assumptions of the particulate model of matter. 2

19. Define an element. Give two examples. 2

20. Define solute, solvent and solution. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how force can change the shape of an object. Give two examples. 3

22. A box of weight 600 N rests on a floor with a base of area 0.5 square metre. Find the pressure. 3

23. Describe an activity to show that particles of matter have spaces between them. 3

24. List three physical properties of metals with an example for each. 3

OR

Describe an activity to show that the solubility of a solid increases with temperature. 3

25. Describe how an electroscope works. 3

OR

Why does a balloon expand when more air is blown in? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the effects of force with five examples. 5

OR

Explain, with examples, how pressure depends on force and area. Calculate the pressure of a 50 N force over 0.25 square metre and over 0.5 square metre. 5

27. Explain the particulate nature of matter with the help of three activities. 5

OR

Compare elements, compounds and mixtures with examples in a table of five points. 5

END OF QUESTION PAPER — Total: 50 Marks

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